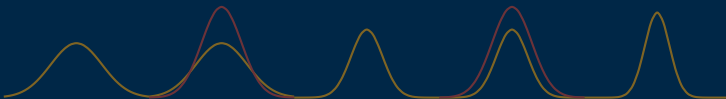


# Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

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Masters dissertation ▷ Thiago Tomás de Paula

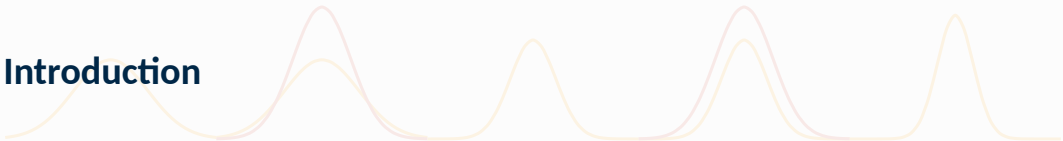


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# Introduction



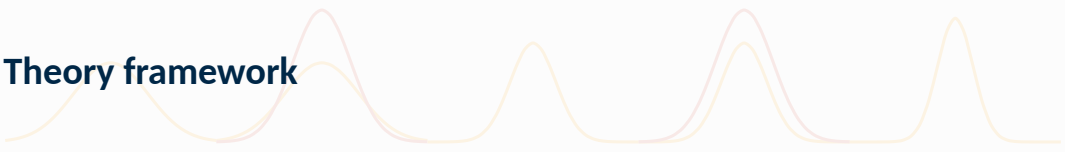








**Theory framework**

The text "Theory framework" is positioned on the left side of the slide. To its right, there are two decorative, wavy lines that span across the middle of the slide. One line is a light yellow color, and the other is a light pink color. They overlap and create a rhythmic, undulating pattern.



### Prediction forward model

The prediction forward model, or forward model, is the underlying mapping between a model's parameters, controls and initial states to its observable features.

Denoting by  $\mathbf{m}$  the vector of model parameters, and by  $\mathbf{u}$  the set of controls applied on the field, and by  $\mathbf{d}$  the reservoir production within the time frame of the controls, one can write

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}; \mathbf{x}_0) \quad (1)$$

with  $\mathbf{x}_0$  the initial value of reservoir states (usually omitted).





## Data features

In this work, the features inside  $\mathbf{d}$  are the following well measurements:

- ▷ oil and water production rates (OPR and WPR) [ $\text{m}^3/\text{day}$ ]
- ▷ water injection rate (WIR) [ $\text{m}^3/\text{day}$ ]
- ▷ bottomhole pressure (BHP) [Bar]
- ▷ water cut percentage (WCTP) [%]

## Field oil production total

Field oil production total (FOPT, SI unit of  $\text{m}^3$ ) is also relevant to this work, despite not being part of  $\mathbf{d}$ .



The key assumption behind the forward model is that it is deterministic. Implementation wise,  $g$  consists of evaluating a set of equations at each gridblock of the model.

Each evaluation evolves the model a little further in time, until the scheduled time period is appreciated, concluding the process.

Black-oil equations are the most common implementations of the forward model [2] due to their fair simplicity and accuracy. The fluid is assumed composed of three liquid phases, water, oil and gas.



A given phase  $\alpha$ 's component mass  $A_\alpha$  satisfies the conservation equation

$$\frac{\partial}{\partial t} (\phi_{\text{ref}} A_\alpha) + \nabla \cdot \mathbf{u}_\alpha + q_\alpha = 0, \quad (2)$$

where

$$A_w = m_\phi b_w s_w, \quad \mathbf{u}_w = b_w \mathbf{v}_w, \quad (3a)$$

$$A_o = m_\phi (b_o s_o + r_{og} b_g s_g), \quad \mathbf{u}_o = b_o \mathbf{v}_o + r_{og} b_g \mathbf{v}_g, \quad (3b)$$

$$A_g = m_\phi (b_g s_g + r_{go} b_o s_o), \quad \mathbf{u}_g = b_g \mathbf{v}_g + r_{go} b_o \mathbf{v}_o. \quad (3c)$$

and the phase fluxes are given by Darcy's law:

$$\mathbf{v}_\alpha = -\lambda_\alpha \mathbf{K} (\nabla p_\alpha - \rho_\alpha \mathbf{g}). \quad (4)$$

Finally, the following closure equations are imposed:

$$s_w + s_o + s_g = 1, \quad p_{c,ow} = p_o - p_w, \quad p_{c,og} = p_o - p_g. \quad (5)$$



Let  $\mathbf{d}$  denote the true production data vector of a reference reservoir across time, free of observation error. Data is separated into a history period and a validation period, so that

$$\mathbf{d} = \begin{bmatrix} \mathbf{d}_h \\ \mathbf{d}_v \end{bmatrix} \quad (6)$$

with  $\mathbf{d}_h$  the ground-truth history data, and  $\mathbf{d}_v$  the ground-truth validation data.

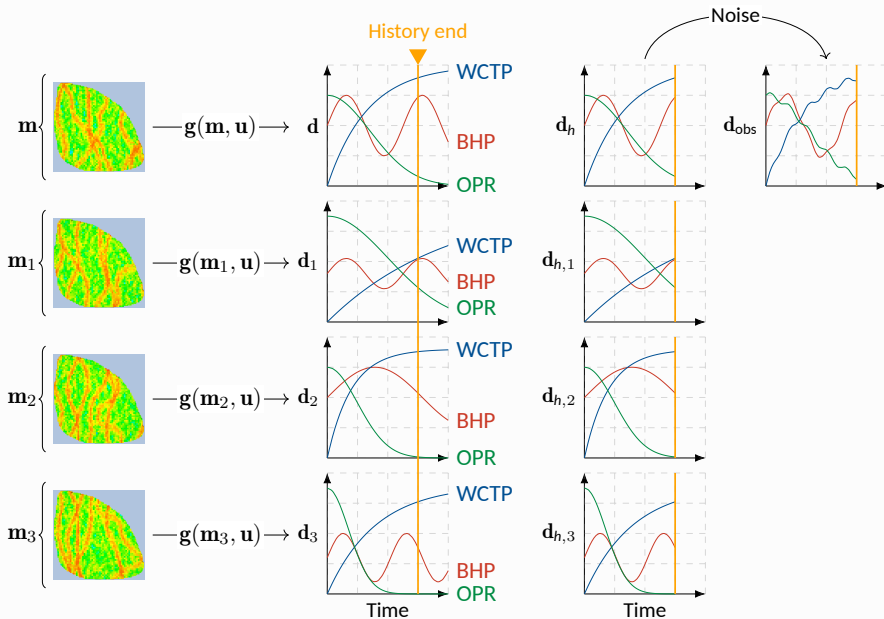
Ground-truth data come from the forward model,

$$\mathbf{d} = \mathbf{g}(\mathbf{m}, \mathbf{u}). \quad (7)$$

and observed data, from a corruption of  $\mathbf{d}_h$ :

$$\mathbf{d}_{\text{obs}} = \mathbf{d}_h + \mathcal{N}(\mathbf{0}, \mathbf{C}_e), \quad (8)$$

where  $\mathbf{C}_e$  is the noise covariance matrix.





DSI-ESMDA can be conceptualized as a Bayesian approach to the forecasting problem, aiming to estimate the underlying conditional probability distribution of predictions given the reference data.

## The DSI-ESMDA problem

Find the posterior distribution

$$p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}}) \propto \mathcal{L}(\mathbf{d}_{\text{obs}} \mid \mathbf{d}^*)p(\mathbf{d}^*). \quad (9)$$

where  $\mathbf{d}^*$  maximizes  $p(\mathbf{d}^* \mid \mathbf{d}_{\text{obs}})$ .

The algorithm returns samples from this posterior distribution, outputting a range of values rather than a single value. While this makes evaluating the algorithm less straightforward, it allows for quantification of uncertainty.



Ensemble Smoothers share this approach at a conceptual level, but they perform a single maximization. This leads to poor matches to real data if assumptions on the likelihood  $\mathcal{L}$  are not met.

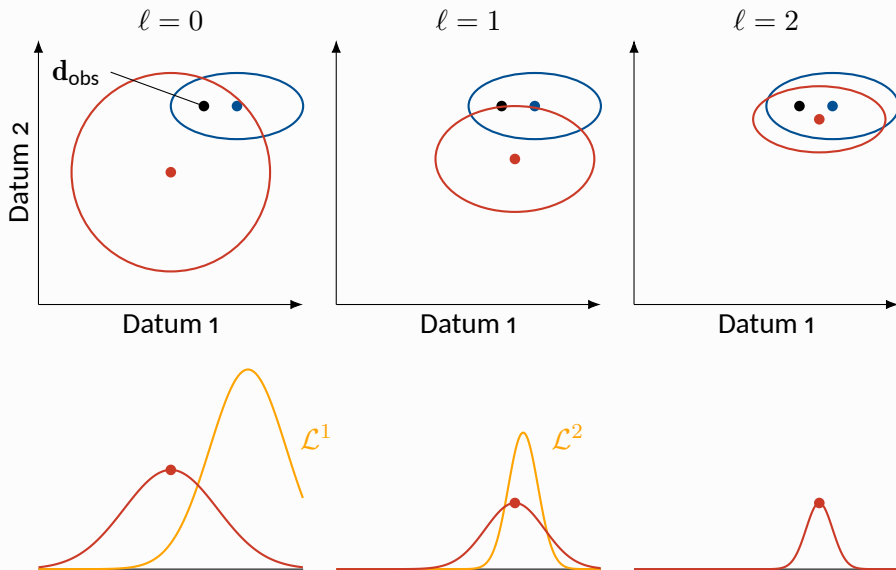
Multiple Data Assimilation (MDA) mitigates this problem by making  $N_a$  optimizations with resampled reference data:

$$p^1(\mathbf{d}^* | \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^1 | \mathbf{d}^*)p(\mathbf{d}^*) \quad (10)$$

$$p^2(\mathbf{d}^* | \mathbf{d}_{\text{obs}}^2, \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^2 | \mathbf{d}^*)p^1(\mathbf{d}^* | \mathbf{d}_{\text{obs}}^1) \quad (11)$$

$$\vdots \quad (12)$$

$$p^{N_a}(\mathbf{d}^* | \mathbf{d}_{\text{obs}}^{N_a}, \dots, \mathbf{d}_{\text{obs}}^1) \propto \mathcal{L}(\mathbf{d}_{\text{obs}}^{N_a} | \mathbf{d}^*)p^{N_a-1}(\mathbf{d}^* | \mathbf{d}_{\text{obs}}^{N_a-1}, \dots, \mathbf{d}_{\text{obs}}^1) \quad (13)$$







MDA makes the following suppositions on the likelihoods:

- $\mathcal{L}^\ell$  is Gaussian
- $\mathcal{L}^\ell$  has covariance  $\mathbf{C}_e^\ell = \alpha^\ell \mathbf{C}_e$ ,  $\alpha^\ell > 0$

This leads to

$$\mathcal{L}^\ell(\mathbf{d}_{\text{obs}}^\ell \mid \mathbf{d}^*) \propto \exp \left( -\frac{1}{2} (\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*)^\top \frac{\mathbf{C}_e^{-1}}{\alpha^\ell} (\mathbf{d}_{\text{obs}} - \mathbf{H}\mathbf{d}^*) \right), \quad (14)$$

$\mathbf{H}$  a matrix that extracts  $\mathbf{d}_h^*$  from  $\mathbf{d}^*$ .

## Equivalence to ES

For Gaussian likelihoods, MDA is equivalent to ES if

$$\sum_{\ell=1}^{N_a} \frac{1}{\alpha^\ell} = 1, \quad (15)$$

highlighting the importance of having inflation factors on the covariance.



The priors are also supposed Gaussian:

$$p^\ell(\mathbf{d}^* \mid \mathbf{d}_{\text{prior}}^\ell) \propto \exp \left( -\frac{1}{2} (\mathbf{d}^* - \mathbf{d}_{\text{prior}})^\top (\mathbf{C}_p^\ell)^{-1} (\mathbf{d}^* - \mathbf{d}_{\text{prior}}) \right), \quad (16)$$

where

- ▷  $\mathbf{d}_{\text{prior}}^\ell$  is the prediction with highest belief of occurring prior to assimilating  $\mathbf{d}_{\text{obs}}^\ell$
- ▷  $\mathbf{C}_p^\ell$  is the prior covariance matrix (i.e.,  $\mathbf{d}^* \sim \mathcal{N}(\mathbf{d}_{\text{prior}}^\ell, \mathbf{C}_p^\ell)$ )



The maximum *a posteriori* (MAP) estimate of the  $\ell$ -th assimilation becomes

$$\mathbf{d}_{\text{map}}^{\ell+1} = \mathbf{d}_{\text{map}}^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}}^{\ell} - \mathbf{d}_{h,\text{map}}^{\ell}), \quad (17a)$$

$$\mathbf{K}^{\ell} = \mathbf{C}_p^{\ell} \mathbf{H}^{\top} (\mathbf{H} \mathbf{C}_p^{\ell} \mathbf{H}^{\top} + \alpha^{\ell} \mathbf{C}_e)^{-1} \quad (17b)$$

with  $\mathbf{d}_{\text{map}}^0 = \mathbf{d}_{\text{prior}}^0$ .

Using an ensemble of  $N_e$  model realizations, one can estimate the prior covariances with  $\Delta \mathbf{D} = [\mathbf{d}_1 - \bar{\mathbf{d}} \quad \dots \quad \mathbf{d}_{N_e} - \bar{\mathbf{d}}]$ :

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + \mathbf{K}^{\ell}(\mathbf{d}_{\text{obs}}^{\ell} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (18a)$$

$$\mathbf{K}^{\ell} = \Delta \mathbf{D}^{\ell} \Delta \mathbf{D}_h^{\ell} (\Delta \mathbf{D}_h^{\ell} \Delta \mathbf{D}_h^{\ell} + \alpha^{\ell} \mathbf{C}_e)^{-1}. \quad (18b)$$

$\mathbf{d}_{\text{obs}}^{\ell} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell}$ , with  $\mathbf{e}_j \sim \mathcal{N}(\mathbf{0}, \mathbf{C}_e)$ , is a way of sampling  $\mathbf{d}_{\text{obs}}^{\ell} \sim \mathcal{N}(\mathbf{d}_{\text{obs}}, \alpha^{\ell} \mathbf{C}_e)$ .



Covariance localization, or simply localization, is an *ad hoc* procedure employed to mitigate the effects of small ensembles [1, section 7.5].

The influence a datum has on another datum is weighted in an inverse manner to the distance between the points.

Localization is applied to the Kalman-like gain, so that the base equation becomes:

$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^{\ell} + (\mathbf{R} \circ \mathbf{K}^{\ell})(\mathbf{d}_{\text{obs}} + \sqrt{\alpha^{\ell}} \mathbf{e}_j^{\ell} - \mathbf{d}_{h,j}^{\ell}), \quad (19)$$

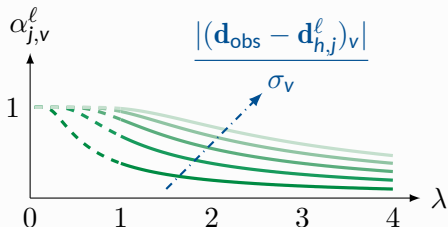
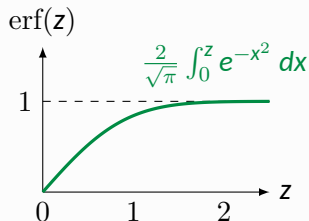


This work introduces gating  $\alpha$  on the innovation vector:

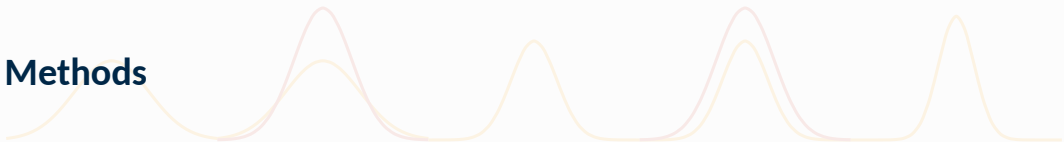
$$\mathbf{d}_j^{\ell+1} = \mathbf{d}_j^\ell + (\mathbf{R} \circ \mathbf{K}^\ell)[\alpha_j^\ell \circ (\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)], \quad (20a)$$

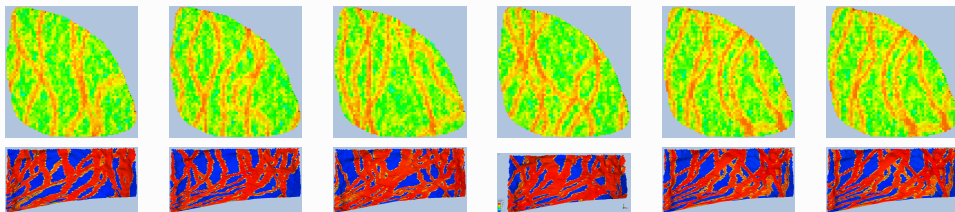
$$\alpha_{j,v}^\ell = \text{erf} \left( \frac{1}{\lambda \sqrt{2}} \frac{|(\mathbf{d}_{\text{obs}} - \mathbf{d}_{h,j}^\ell)_v|}{\sigma_v} \right). \quad (20b)$$

The gating recovers variance loss in the posterior ensemble by diminishing Kalman-like updates on reasonably matched data. The smaller the mismatch of a measurement relative to the noise at that datum, the smaller the update.



## Methods

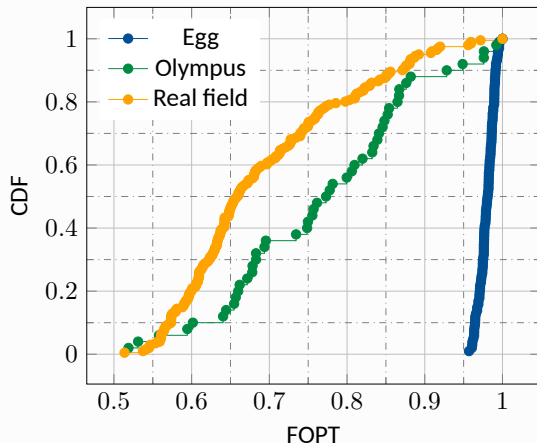




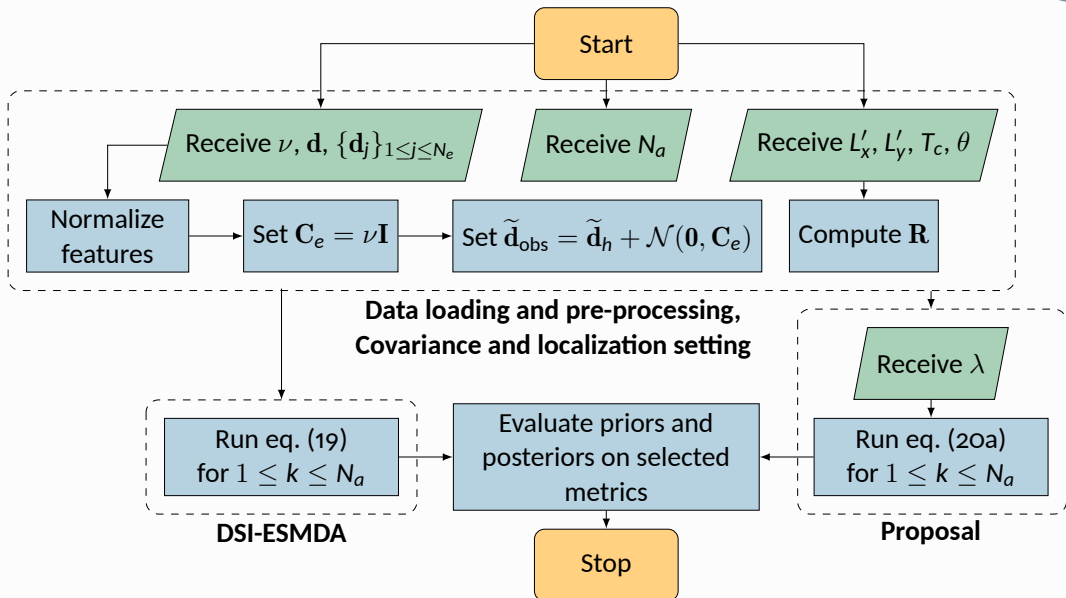
| Property                | Egg                   | Olympus                  | Real field | Unit  |
|-------------------------|-----------------------|--------------------------|------------|-------|
| Rock compressibility    | 0.0                   | $1.42 \times 10^{-10}$   | -          | 1/Pa  |
| Water compressibility   | $1.0 \times 10^{-10}$ | $3.97 \times 10^{-10}$   | -          | 1/Pa  |
| Oil dynamic viscosity   | $5.0 \times 10^{-3}$  | $2.8-3.5 \times 10^{-3}$ | -          | Pa·s  |
| Water dynamic viscosity | $1.0 \times 10^{-3}$  | $0.398 \times 10^{-3}$   | -          | Pa·s  |
| Well diameter           | 20                    | 19                       | -          | cm    |
| Wells                   | 8(P) 4(I)             | 11(P) 7(I)               | 16(P) 8(I) | -     |
| Active cells            | 18,533                | 192,750                  | -          | -     |
| Cell size               | $8 \times 8 \times 4$ | $50 \times 50 \times 3$  | -          | m×m×m |
| Ensemble size           | 101                   | 50                       | 201        | -     |



The field oil production total (FOPT) is a measurement commonly used to compare ensembles since ground-truth and realizations are each represented by a single value: how much oil was produced within the period of interest.









Four metrics are considered on the evaluation of posterior/prior distributions [1, subsection 9.4.2]: history observation coverage, validation observation coverage, mean squared error, and data-mismatch.

Denoting an ensemble by  $\mathbf{Y} = [\mathbf{y}_1 \ \cdots \ \mathbf{y}_{N_e}]$ , these metrics are defined as:

$$\text{OC}_h(\mathbf{Y}) = \frac{1}{N_h} \sum_{i=1}^{N_h} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{h,i} \leq \max(\mathbf{Y})_i) \quad (21)$$

$$\text{OC}_v(\mathbf{Y}) = \frac{1}{N_v} \sum_{i=1}^{N_v} \mathbb{1} (\min(\mathbf{Y})_i \leq \mathbf{d}_{v,i} \leq \max(\mathbf{Y})_i) \quad (22)$$

$$\text{MSE}(\mathbf{Y}) = \frac{1}{N_h} (\mathbf{d}_h - \bar{\mathbf{y}})^\top \mathbf{C}^{-1} (\mathbf{d}_h - \bar{\mathbf{y}}), \quad \mathbf{C} = \Delta \mathbf{Y} \Delta \mathbf{Y}^\top + \mathbf{C}_e \quad (23)$$

$$\mathcal{O}(\mathbf{y}_j) = \frac{1}{2N_h} (\mathbf{d}_h - \mathbf{y}_{j,h})^\top \mathbf{C}_e^{-1} (\mathbf{d}_h - \mathbf{y}_{j,h}) \quad (24)$$

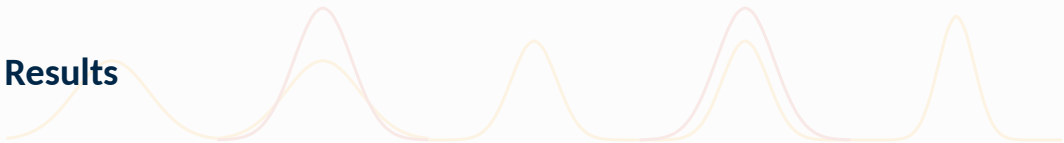


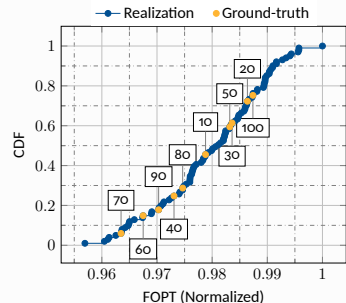
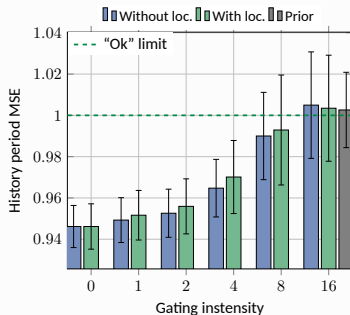
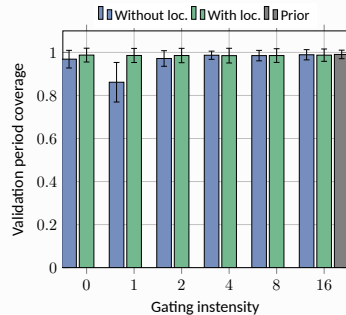
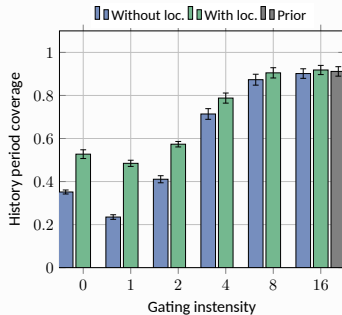
| Reservoir  | $N_e$ | Number of time steps | Time step size   | Simulation life       | History split |
|------------|-------|----------------------|------------------|-----------------------|---------------|
| Egg        | 100   | 120                  | 30 days          | 3600 days (9.8 years) | 60%           |
| Olympus    | 49    | 20                   | 1 calendar year  | 7305 days (20 years)  | 60%           |
| Real field | 200   | 510                  | 1 calendar month | 12785 days (35 years) | 60%*          |

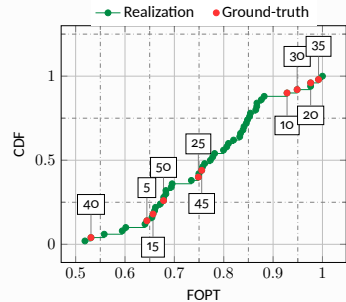
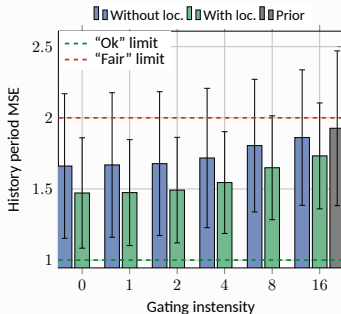
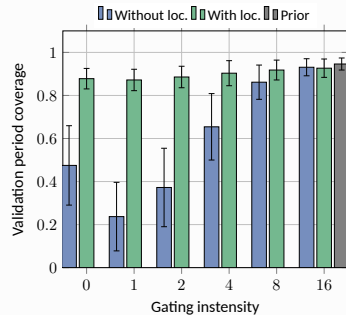
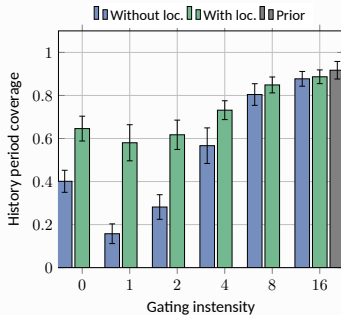
| Reservoir  | Critical length in x | Critical length in y | Critical time | Anisotropy |
|------------|----------------------|----------------------|---------------|------------|
| Egg        | 100                  | 100                  | 3000 days     | 0°         |
| Olympus    | 100                  | 100                  | 7200 days     | 0°         |
| Real field | -                    | -                    | 10500 days    | 0°         |

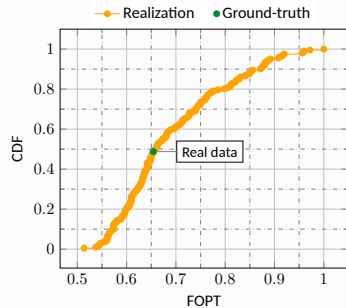
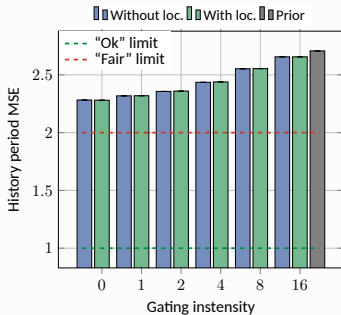
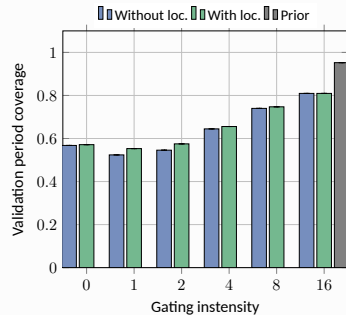
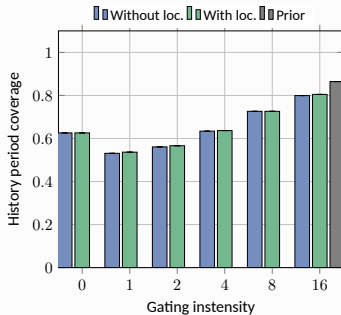
| Reservoir  | Well features used |          | Number of wells |          | Total features           |
|------------|--------------------|----------|-----------------|----------|--------------------------|
|            | Producer           | Injector | Producer        | Injector |                          |
| Egg        | OPR, WPR           | BHP      | 4               | 8        | 16                       |
| Olympus    | OPR, WPR           | WIR      | 11              | 7        | 29                       |
| Real field | BHP, WCTP          | BHP      | 16              | 8        | 39 (a WCTP was constant) |

## Results











## OC, MSE and mismatch increase

Increase in gating intensity tended to increase observation coverage, mean squared error, and data mismatch, albeit with case-dependent sensitivities.

- ▶ Gating reduces posterior variance loss, so increase in coverages, but also MSE and mismatch, are expected as gating intensifies.
- ▶ Egg standard deviations are lesser than Olympus on two accounts: reservoir complexity, and ground-truth representativeness. Olympus' ground-truths were somewhat biased.

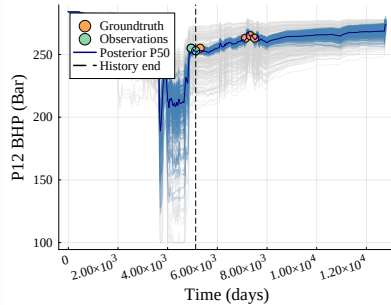
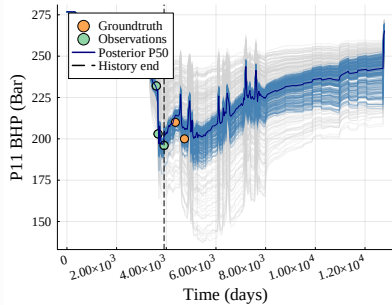
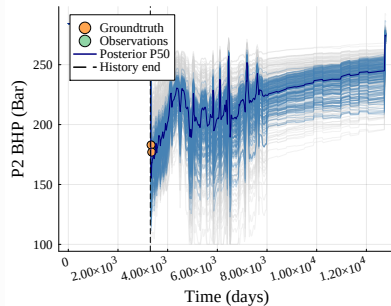
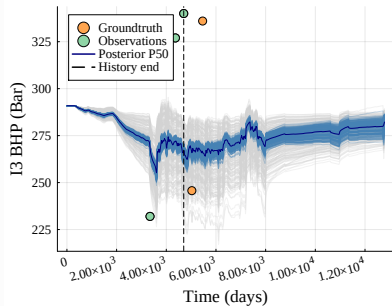




## Mixed effects of localization

Localization yielded mixed results: for the Egg and real field cases, effects on coverage and MSE were minimal, especially for intense enough gatings, while the effect in the Olympus case was drastic.

- ▷ **Egg case:** narrow FOPT range and roughly uniform sampling of ground-truths leave no ground-truth unrepresented, reduced dimensionality help with covariance estimation.
- ▷ **Olympus case:** much more complex while also having a much smaller ensemble and a lot more features, aggravating covariance estimation. Localization was conceived exactly for such situations, so it is not surprising it was effective here.
- ▷ **Real field case:** peak complexity and dimensionality, but also ensemble size. Localization may have inadequately adjusted, and some features are not well-represented.





## Mild and small gating intensities versus standard DSI-ESMDA

Mild gating intensities (2 in the Egg and Olympus cases, 4 in the real field case) produced similar results to the standard DSI-ESMDA for same localization configuration, while small gating (1 for the Egg case, 1 and 2 for the Olympus and real field cases) had worse coverages than the standard DSI-ESMDA.

- ▷ Gating intensity “reverts” the posterior to the prior. If the posterior given by the standard DSI-ESMDA is somewhere in that continuum, then it stands to reason that Gated DSI-ESMDA could approximate it simply by tuning the gating intensity.
- ▷ Since Gated DSI-ESMDA does away with observation data resampling, small gatings possibly underestimate uncertainty (i.e., likelihood covariance) at least when compared to standard DSI-ESMDA, producing posteriors that are not well-matched to data.



- ▷ This work introduced Gated DSI-ESMDA, an extension of DSI-ESMDA with a nonlinear innovation gating mechanism that increases posterior variance the more intense the gating.
- ▷ Observation coverages of the posterior increased with the gating intensity for all case studies, albeit under different sensitivities.
- ▷ Small gatings tend to underestimate uncertainty, while mild gatings produced similar results to standard DSI-ESMDA.
- ▷ The gating mechanism in of itself can be easily adapted into model-space methods such as ES-MDA.
- ▷ Gated DSI-ESMDA remains susceptible to poor feature selection and localization tuning, as well as insufficient ground-truth representativeness, high data dimensionality, and reduced ensemble size. Future research should address these problems.

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- [1] Alexandre Anozé Emerick. *Ensemble Data Assimilation Applied to Geological Reservoir Models*. 1st ed. Rio de Janeiro, RJ, Brazil: Universidade Petrobras, June 2025. ISBN: 978-65-88763-29-2.
- [2] Atgeirr Flø Rasmussen et al. "The Open Porous Media Flow reservoir simulator". In: *Computers & Mathematics with Applications* 81 (2021). Development and Application of Open-source Software for Problems with Numerical PDEs, pp. 159–185. ISSN: 0898-1221. DOI: <https://doi.org/10.1016/j.camwa.2020.05.014>.

# Gated DSI-ESMDA

Reducing variance loss in posteriors

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August 4th, 2026

*“It’s hard to make predictions, specially about the future”*

— Yogi Berra